



A System Analysis and Design of Marketing Strategy for Improving Pineapple Agritourism

Kartika Trianita
Bogor Agricultural University
Bogor, Indonesia
kartika_trianita@apps.ipb.ac.id

Taufik Djatna
Bogor Agricultural University
Bogor, Indonesia
taufikdjatna@ipb.ac.id

Anas Miftah Fauzi
Bogor Agricultural University
Bogor, Indonesia
amfauzi@apps.ipb.ac.id

ABSTRACT

Marketing strategy is the most important factor to increase the visitor's number to an agritourism. Accordingly, the selection of the best marketing strategy is needed so it provides the maximum result. Failure of strategy implementation can cause income loss for agritourism operational. Hence, adequate system analysis and business process design are needed to reduce the possibility of failure. This paper aims to analyze the requirements of the system and to design the business process including identify and determine visitor preferences in agritourism facilities and determine association rules of the priority marketing strategy for improving pineapple agritourism. Analytical system entity construction was used to describe the requirements of the system, the Unified Modeling Language and Business Process Model and Notation 2.0 were used to design the business process. Due to the limited marketing budget, the top eight most interesting facilities based on visitor preferences were determined by RELIEF-F algorithm. Then, the actionable marketing rules were determined by ARM algorithm. The result of system analysis shows that the system requires facilities and visitor preferences as inputs, agritourism operational and visitor as stakeholders, to result in an output of association rules as improved marketing strategies. From the design, we obtained the top eight facilities are camping ground, photo spot, pineapple field tour, pineapple factory tour, culture attraction, bicycle track, gazebo, and playground. ARM results association rules design is composed of five facilities: pineapple factory, pineapple field, gazebo, playground, and photo spot. From the result, we can conclude that the agritourism operational should consider placing the facilities close to each other according to obtained association rules.

CCS Concepts

• Computing methodologies → Feature selection

Keywords

agritourism; ARM; marketing strategy; pineapple; preference; RELIEF-F

1. INTRODUCTION

Indonesia has many agriculture resources, especially in Java

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Island. However, these resources are still haven't been managed and utilized optimally. Even though, these resources are potential to give added value to their stakeholder in a region. As an example, agritourism based on pineapple is potential to develop in Pemalang Regency as the central production of honey pineapple in Central Java.

Agritourism is defined as all rural enterprises which integrate both a working farm environment and a commercial tourism component [1]. Agritourism can take any form, including a stay in a farmhouse, picking the harvested product, agricultural festivals, field tour, or hayrides. A sustainable agritourism can increase the long-term potential for higher margin on-farm sales of value-added products and services, especially for small farms in crisis [2]. Marketing strategy is needed to promote an agritourism village.

In this paper, we would show the relevance of this work to machine learning discipline. We used data mining techniques to help in formulating a marketing strategy. Recursive Elimination of Feature-F (RELIEF-F) algorithm was used to attribute selection and Association Rule Mining (ARM) algorithm was used to extract interesting correlations in terms of agritourism facilities, frequent patterns, and association among a set of items in the transaction in database dealing with visitor preferences during visits to agritourism facilities.

Marketing strategy is a process by which an organization aligns itself with the target market to be served. In marketing strategy, everyone in the organization is engaged then his or her activities and outputs are focused on the target customer's current and future needs and wants [3]. In strategy formulation, marketing mix (5Ps: Product, Price, Place, Promotion, and People) are needed to be considered [4]. The first P (Product) refers to products and services that are offered in an agritourism. We should consider the target customers are looking for, includes products and services to buy. For the second P (Price), we should consider the customer willingness to purchase our products or services. Suitable price has to meet beneficial of products and services are provided to meet customer's need. Place of agritourism relates to how far your target customers willing to travel and visit the provided facilities and attractions. Thus, the location of the agritourism should cover the reachable area of target customers and adequate facilities are available. Promotion is crucial to be considered, especially in the opening of agritourism or launching new concepts to attract visitors to come. The last that should be considered is People. All of the stakeholders which are involved in agritourism play a role in the development of agritourism. Thus, the needs of every party should be considered to fulfill. Our limitation in this study focus on improving strategy in the place and people aspects. We concern to agritourism operational and visitor as people who play an important role in formulating a marketing strategy. The

place relates to facilities or attractions in agritourism that meet the agritourism operational and visitor needs. Agritourism operational need to grow up the income by increasing the number of visitors. Visitors need to get satisfied facilities and attractions as their expectation. Visitor preferences on facilities are highly important to promote an agritourism based on visitor's needs.

However, implementation of a strategy may fail during formulation stage because of (1) lack of creativity, (2) inadequate or poor analysis, or (3) inability to accept that the business environment has changed [5]. Accordingly, the priority marketing strategy as the optimum strategy to provide a supportive result for implementation. Alternative strategies attempt to develop an agritourism as Strength-Opportunity strategy in addition to other facilities to invite more visitors [6]. First, adequate system analysis is needed to reduce fail possibility in strategy implementation. As the requirements analysis completed, the business process provides direction for design. In this step, we represent the system on a conceptual model as diagrams thus; we can understand the system behavior.

Based on stated facts above, the research objectives are to analyze the requirements of the system and to design the business process including making formulation to identify and determine visitor preferences for agritourism facilities and create marketing rules in improving agritourism. The marketing strategy was formulated by visitor preferences due to the marketing goal is to increase the visitor's number. The strategy was formulated as association rules of the priority marketing strategy for pineapple agritourism.

This research was a study case in Belik Village, Pemalang Regency, Central Java Province, Indonesia as a central production of honey pineapple in Central Java Province. Honey pineapple from Pemalang has a distinctive characteristic compared to other regions. The honey pineapple is sweeter than others because it is planted on the mountain slopes, so the water content is not too much. The economic condition of Belik Village can be improved by increasing the agritourism visitors. A high number of visitors to an agritourism will increase the income of the village population.

2. METHODOLOGY

2.1 Framework

The framework of this research is shown in Figure 1. The requirements of the system's entities were analyzed and constructed in a diagram. The business process was designed using tool SAP PowerDesigner® v16.6.

2.2 Requirements Analysis

Purpose of the system and the expected output was determined by requirements analysis. System entities consist of input (acceptable and unacceptable), stakeholders, roles, mission, and objectives, resources, internal controls/constraints, opportunities, threat, operating constraints were analyzed and described. A system entity was constructed in a diagram [7].

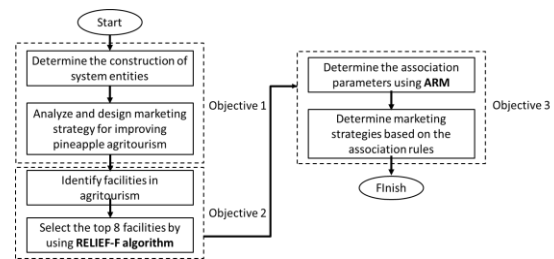


Figure 1 Framework of research

2.3 The Unified Modeling Language (UML)

The artifacts of a software-intensive system can be visualized, specified, constructed, and documented by using UML which is a standard language for writing software blueprints that addressing all the views needed to develop a system. UML diagram is a graphical presentation of a set of system elements between vertices (things) and arcs (relationships). UML includes some diagrams such as use case diagram, sequence diagram, and activity diagram. Use case diagram is a graphical language that shows the system boundaries, interaction and communication relationship between actors. It commonly contains use cases, actors, dependency, generalization, and association relationships. A sequence diagram represents an interaction diagram that emphasizes the time ordering of messages. An activity diagram is used to model the dynamic aspects of a system. It can model the flow of an object as it moves from state to state at different points in the flow of control. It emphasizes the flow of control from activity to activity [8].

2.4 Business Process Modelling (BPM)

A business process model can be presented by using a Business Process Diagram (BPD). BPD contains main tasks for every actor and shows the flow of the business process from actor to actor.

2.5 Business Process Model and Notation (BPMN)

BPMN 2.0 is a graphical language with notations that represents how the business process model is running. Core elements of BPMN are flow objects, connecting object, a swim lane, and artifact. Flow objects contain events, activities, and gateways. Connecting objects contain sequence flow, message flow, and association. Swim lanes contain pool and lanes. Artifacts contain data object, text association, and group [9].

2.6 Formulation Process

Respondent preferences were obtained by utilizing the Google Form Questionnaire which is spread through social media platforms such as Facebook, Instagram, Twitter, and Whatsapp. The data were then used and analyzed by using the Waikato Environment for Knowledge Analysis (WEKA) 3.8 Software.

2.6.1 Recursive Elimination of Feature-F (RELIEF-F algorithm)

RELIEF-F algorithm is used to select the most influential feature. This algorithm has been successfully used in many large subset feature selection tasks, such as selection of informative genes for cancer classification [10], selection of features for emotion recognition [11], selection of the dielectric properties determination of shelf life biscuits [12], and selection of feature for enhancing the decisional DNA-based prediction [13]. The

input is visitor preferences in agritourism facilities from Google Form that has been converted to quantitative score. By using a RELIEF-F algorithm, features were classified and the significance of features were obtained by weighting the features. In this research, respondents gave their preferences in agritourism facilities. RELIEF-F algorithm was used to choose eight out of thirteen potential facilities in agritourism. RELIEF-F algorithm can be formulated as follows [14].

Algorithm RELIEF-F

Input: for each training instance a vector of attribute values and the class value

Output: the vector W of estimations of the qualities of attributes

1. set all weight $W[A] := 0.0$;
2. **for** $l := 1$ **to** m **do begin**
3. randomly select an instance R_i ;
4. find k nearest hits H_j ;
5. **for** each class $C \neq \text{class}(R_i)$ **do**
6. from class C find k nearest misses M_j ;
7. **for** $A := 1$ **to** a **do**
8. $W[A] := W[A] - \sum_{j=1}^k \text{diff}(A, R_i, M_j(C)) / (m \cdot k) +$
9. $\sum_{C \neq \text{class}(R_i)} \left[\frac{P(C)}{1 - P(\text{class}(R_i))} \sum_{j=1}^k \text{diff}(A, R_i, M_j(C)) \right] / (m \cdot k)$;
10. **end**;

2.6.2 Association Rule Mining (ARM)

Eight selected facilities were processed using ARM to get association rules for agritourism facilities procurement as the priority marketing strategy. Stages of association rules are as follows:

1. Calculate the bond value

$$\text{Bond}(AB) = \frac{P(A \wedge B)}{\sum AVB} \times 100\%$$

$$\sum AVB = \text{Conjunctive}$$

$$\sum A \wedge B = \text{Disjunctive}$$

2. Calculate the support value

$$\text{Support}(AB) = \frac{P(A \wedge B)}{\sum n} \times 100\%$$

3. Calculate the confidence value

$$\text{Confidence}(A \rightarrow B) = \frac{P(A \wedge B)}{\text{Conjunctive}(A)} \times 100\%$$

4. Calculate the lift value

$$\text{Lift}(A, B) = \frac{\text{Support}(A \wedge B)}{\text{Support}(A) \times \text{Support}(B)} \times 100\%$$

5. Calculate the conviction value

$$\begin{aligned} \text{Conviction}(A \rightarrow B) &= \frac{1 - \text{Supp}(B)}{1 - \text{Conf}(A \rightarrow B)} \times 100\% \\ &= \frac{P(A)P(\sim B)}{P(A \cap \sim B)} \\ &= \frac{(\text{Support}(A) \times \text{Support}(\sim B))}{\text{Support}(A \rightarrow \sim B)} \times 100\% \end{aligned}$$

6. Calculate the leverage value

$$\begin{aligned} \text{Leverage}(A \rightarrow B) &= P(A \cap B) - (P(A)P(B)) \\ &= (\text{Support}(A \rightarrow B)(\text{Support}(A) \times \text{Support}(B))) \\ &\quad \times 100\% \end{aligned}$$

7. Determine the top 5 association rules

$$\text{Support value} \times \text{Conf value}$$

8. Formulate association rules into mathematical models

$$R_n = \{A_1^n\} \rightarrow \{Y^n\}$$

This research used the a priori algorithm to form association rules. A priori algorithm is used for mining frequent item set using Boolean association rules [15]. The algorithm is formulated as below [16]:

$k = 1$

$F_k = \{i | i \in I \wedge \sigma(\{i\}N \times \text{minsup})\}$
 {Find all frequent 1 – itemsets}

repeat

$k = k + 1$

$C_k = \text{apriori} - \text{gen}(F_{k-1})$
 {Generate candidate itemsets}

for each transaction $t \in T$ **do**

$C_t = \text{subset}(C_k, t)$

{Identify all candidates that belong to t }

for each candidates itemset $c \in C_t$ **do**

$\sigma(c) = \sigma(c) + 1$

{Increment support count}

end for

$F_k = \{c | c \in C_k \wedge \sigma(c) \geq N \times \text{minsup}\}$

{Extract the frequent k – itemsets}

until $F_k = \emptyset$

$\text{Result} = \cup F_k$

2.7 Verification and Validation

IEEE defines verification and validation is “the process of determining whether the requirements or components of a system are complete and correct, the products of each development phase fulfill the requirements or conditions imposed by the previous phase, and the final system or component complies with specified requirements” [17]. In this research, use case diagram, BPD diagram, and BPMN 2.0 should show zero error(s) and zero warning(s) results to indicate that the object-oriented and business process are correct. For the design, we verify whether the mathematical function result as expected according to its purpose or not. The objectives of validation include evaluating and assessing system design solution during system development to determine whether it will satisfy its user's operational needs. The user could be the external user or system developer, engineer, designer, or tester. Moreover, validation includes conducting field exercises with the actual users who have been trained to operate the deliverable system, product, or service to coordinate missions [18]. In this research, the designer as user evaluated and assessed whether the system satisfies the designer's operational needs or not.

3. RESULTS AND DISCUSSION

3.1 System Definition

The construction of system analysis and design in marketing strategy to improve pineapple agritourism is shown in Figure 2. Based on the diagram, the system is focused on two stakeholders, Agritourism operational and Visitor. Meanwhile, the system input consists of Acceptable and Unacceptable input, which both include Facilities and Visitor preferences. However, the acceptable input considers potential facilities and visitor, which is different from the unacceptable input that concern non-potential facilities and visitors. The system also requires several resources such as agritourism area, visitor preferences database, and financial budget. In term of visitor preferences, the questionnaire can be utilized to control visitor's criteria by applying several classifications such as age, gender, or hometown. On the other side, improper visitor preferences data and lack of marketing budget could be potential threats to these

systems. The preference is improper when information from the questionnaire is inaccurate due to unauthentic data obtained from unreliable respondents. Strategies that could be implemented on this occasion depend on the marketing budget. For instance, if the cost exceeds the budget, the implementation of the main strategy could be impossible. Thus, the strategies design and execution should follow the cost constraint, from which the output of strategy selection could be affected.

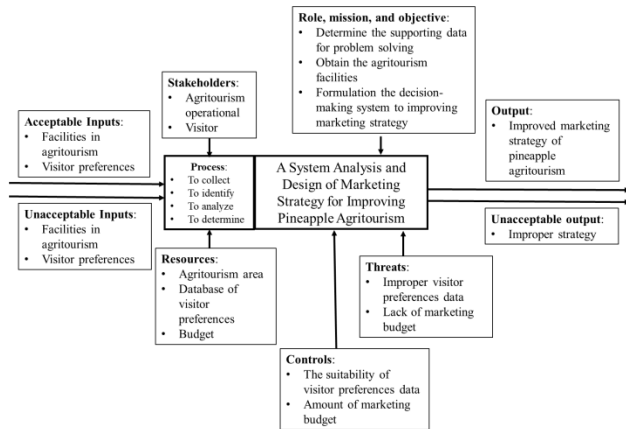


Figure 2. Construction of system's entities

3.2 UML

The use case diagram is shown in Figure 3. Agritourism operational is the stakeholder who responsible for determining the KPI of marketing, identify agritourism facilities and criteria of the respondent to obtain preferences in determining the marketing strategy. The visitor is the stakeholder who responsible in provide preferences as the basis for strategies formulation that conducted by agritourism operational. Visitors include people who have visited Belik Village, people who have visited the other agritourism villages, or people who interested in visiting agritourism village.

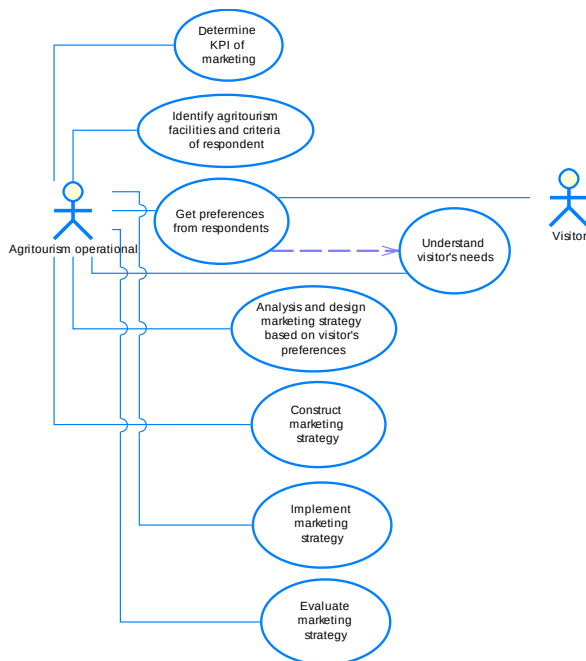


Figure 3. Use case diagram

3.3 BPM

The BPD diagram is shown in Figure 4. BPD shows that agritourism operational holds the lead role as the stakeholder who formulates the strategy. Visitors just conduct one critical task that is filled in the questionnaires to give preferences, but these are important for constructing marketing strategy by agritourism operational.

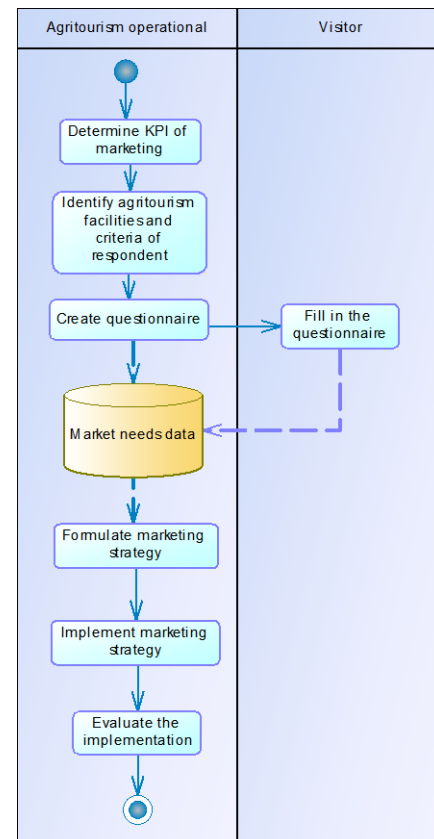


Figure 4. BPD diagram

3.4 BPMN

BPMN 2.0 shows how the business process is running. There are three critical tasks in formulating a marketing strategy for improving pineapple agritourism. First, the business process is started from agritourism operational who determine the KPI of the marketing strategy. Due to the objective of the marketing strategy is to increase the number of visitors, then the marketing KPI could be determined by measuring the achievement of the number of visitors from the target. Second, the agritourism operational analyzes the agritourism facilities which is interested to visit and analyzes the criteria of respondents. Analysis of potential facilities could be done by literature studies of facilities in other similar agritourism. The criteria of respondents could be classified by gender, age, or hometown. The age could be classified into child age, young age, middle age, and senior age. Hometown could be classified by the distance from agritourism location, like Central Java Province as the nearest area, West Java Province and East Java Province as the rather near areas, and other provinces. In this research, we classified the data by gender. Then, the agritourism operational creates a questionnaire and distributes it to respondents according to the criteria. Finally, in the most critical task (Figure 5), the data of questionnaires are processed using RELIEF-F algorithm and followed by ARM

algorithm to provide improved marketing strategies. After getting the marketing strategies, we can analyze what the best facilities combination, thus the agritourism operational can make a program to renovate or add new facilities.

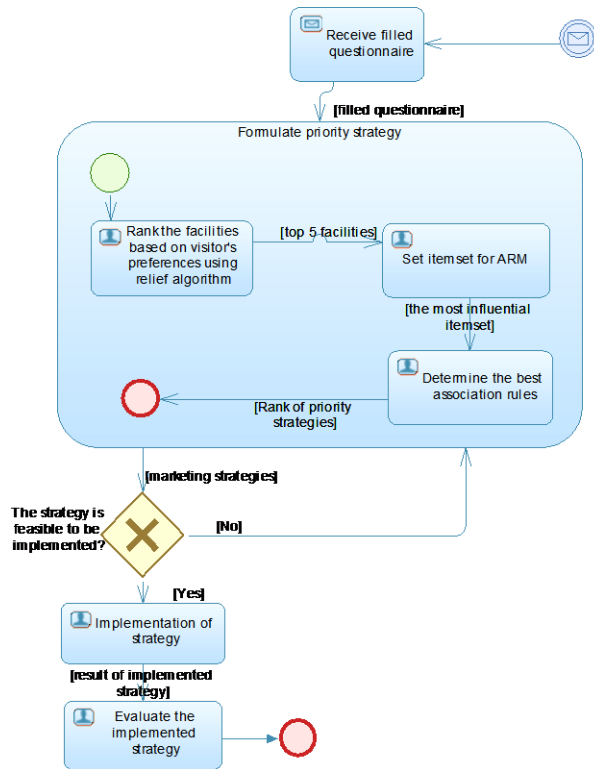


Figure 5. The critical task in formulating a marketing strategy

3.5 Feature Selection

89 respondents filled and sent back the questionnaire. The classification of the questionnaire's data is based on the respondent gender, male and female. By using RELIEF-F algorithm, we obtained the rank of facilities based on visitor preferences. Rank 1 is the most interesting facility and rank 13 is the least interesting facility. The most interesting facility is a camping ground followed by photo spot, pineapple field tour, pineapple factory tour, culture attraction, bicycle track, gazebo, playground, Wi-Fi, souvenir shop, outbound, café and homestay (Table 1). The agritourism operational shall focus on the development of these facilities to increase the visitor's number. The big 8 facilities are continued in association rules formulation to get the best marketing strategy.

Table 1. The result of RELIEF-F algorithm

Attribute	Weight	Rank
Camping ground	0.06292	1
Photo spot	0.05730	2
Pineapple field tour	0.05169	3
Pineapple factory tour	0.04607	4
Culture attraction	0.02921	5
Bicycle track	0.02360	6
Gazebo	0.01573	7

Playground	0.01348	8
Wi-Fi	0.00337	9
Souvenir shop	0.00225	10
Out bond	0.00225	11
Café	-0.00112	12
Homestay	-0.01124	13

3.6 Association rules

As a marketing strategy, ARM resulted in association rules that are constructed by five facilities: pineapple factory tour, pineapple field tour, gazebo, playground, and photo spot. The 10 best combinations of facilities in minimum support count 85% and confidence more than 90% are shown in Table 2.

From the association rules, the first datum means that there are 90% visitors who will come to both of pineapple factory and gazebo. Then, we confidence 100% that if a visitor comes to the pineapple factory, he will also come to the gazebo. As well as the next data for other association rules. From the data, we can suggest that the position of the pineapple factory, gazebo, playground, photo spot, and pineapple field should be close together so the possibility to visit by visitors could be increased.

Table 2. Association rules

Antecedent		Consequent	Support count	Confidence
I F	Pineapple factory	Gazebo	90%	100%
	Playground	Gazebo	88%	100%
	Pineapple factory and photo spot	Gazebo	88%	100%
	Pineapple factory and playground	Gazebo	87%	100%
	Pineapple field	Pineapple factory	85%	100%
	Pineapple field	Gazebo	85%	100%
	Pineapple field and Gazebo	Pineapple factory	85%	100%
	Pineapple field and pineapple factory	Gazebo	85%	100%
	Pineapple field	Pineapple factory and Gazebo	85%	100%
	Photo spot and playground	Gazebo	85%	100%
		T H E N		

3.7 Verification and validation

BPMN resulted in 0 error(s), 0 warning(s). It means the object-oriented and business process model is correct. The design was verified since the mathematical function result as expected according to its purpose. The rank of facilities as shown in Table 1 is a result of the mathematical function in Section 2.6.1 and association rules as shown in Table 2 is a result of the mathematical function in Section 2.6.2.

Validation was done by evaluated and assessed by the designer as the user and it satisfied the designer's operational needs since the result of UML and BPMN meet the system requirements and expected output. Moreover, the validation process could be continued by agritourism operational as the actual user of this system and assessed whether the system could satisfy the agritourism operational needs.

4. CONCLUSION

This work concludes that from the system analysis, the system requires two inputs consists of facilities and visitor preferences, two stakeholders consist of agritourism operational and visitors to results an output, association rules as the marketing strategy for improving pineapple agritourism. From the design, we obtained the rank of top eight facilities based on visitor preferences are camping ground, photo spot, pineapple field tour, pineapple factory tour, culture attraction, bicycle track, gazebo, and playground. Actionable marketing rules are composed of five facilities; pineapple factory, pineapple field, gazebo, playground, and photo spot. As a marketing strategy, agritourism operational should consider placing the facilities close to each other according to obtained association rules.

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